

SMART INDIA HACKATHON 2024



SMART
HACKATHON
2024



- Problem Statement ID – 25129
- Problem Statement Title – Automating GPO Customization and Deployment Based on CIS Guidelines
- Theme – Smart Education
- PS Category – Software
- Team ID – 27113
- Team Name – Cannon Crew



CIS-GPO AUTOMATOR FOR AIR-GAPPED SECURITY



Detailed Explanation of the Proposed Solution

- **Desktop App / CLI:** A lightweight tool capable of automating GOP Customization based on CIS Guidelines
- **Local AI:** We use Local AI and LLMs to automate the customization process.
- **PDF Upload:** Users upload CIS guidelines as PDF, accessed by the LLM via RAG pipeline.
- **Easy Testing:** Our in-house suite of tools ensures the GOP's are enabled properly
- **Import and Export Settings:** GOP settings can be easily imported and exported in any file format.
- **Saving and Documentation:** Save Group configurations for future use with rich documentation.

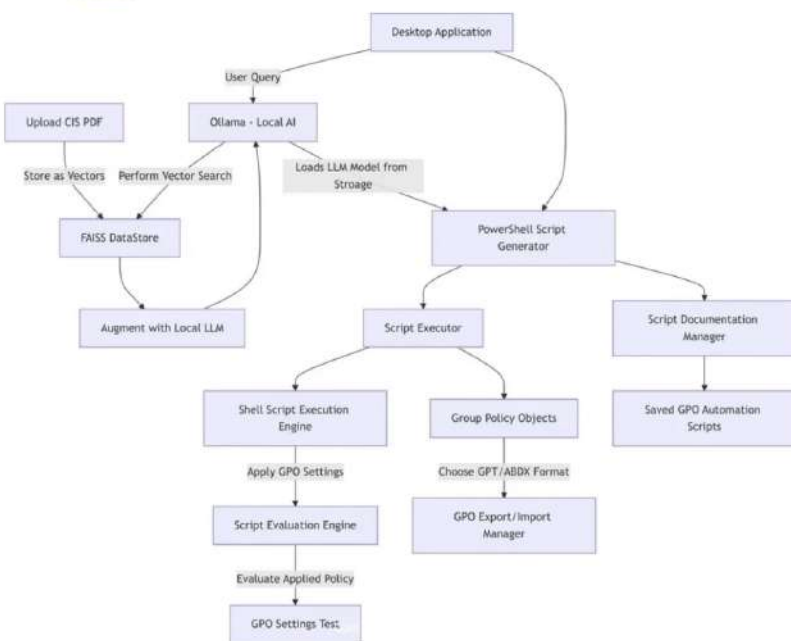
How It Addresses the Problem?

- **Automation:** Turn CIS guidelines into PowerShell, reducing manual work for complex security setups.
- **Air-gapped Friendly:** Runs locally, ideal for high-security environments without internet. Even the AI component.
- **Customizable:** Tailors GPO settings to user needs while adhering to CIS standards.
- **Consistent:** Exports/imports policies for easy application across systems.
- **Validated:** Includes security tests to ensure compliance.
- **Documented:** Maintains clear records of applied security measures.

Innovation and Uniqueness of the Solution

- **Local LLM Execution:** Runs LLMs offline using Ollama for advanced AI capabilities in air-gapped environments.
- **Integrated RAG Pipeline:** Uses FAISS to efficiently process CIS guidelines.
- **End-to-End Automation:** Streamlines security processes from guideline ingestion to script application.
- **Adaptive Learning:** Processes new guidelines through PDF uploads to stay current with security standards.
- **Offline AI-Powered Security:** Provides a unique cybersecurity solution for sensitive environments by combining local AI processing, policy management, and system hardening.

TECHNICAL APPROACH



• ARCHITECTURE

- Chain of LLM agents with access to an admin shell.
- Use of power shell scripts to automate GPO application.
- LLM (Mistral) generates Power Shell script using CIS guidelines stored in FAISS (RAG Database)
- Admin Shell executes Powershell Script and implements GPO rules
- GPO rules are tested with in- house testing and analysis tools.

• COMPONENTS :

- Script Engine: Automates GPO generation and customization.
- Database: Stores and manages CIS guidelines, and user-defined settings.
- Deployment Module: Deploys the GPOs on air-gapped/ standalone systems via admin shell.
- Auditing/ Testing Tool: Verifies system compliance post-deployment.

TECHNOLOGY STACK



ANALYSIS OF THE FEASIBILITY OF THE IDEA

Technical and Operational Feasibility: Proven scripting techniques and tools like Ollama allow for full remote management and 100% offline functionality, ideal for secure, high-security environments.

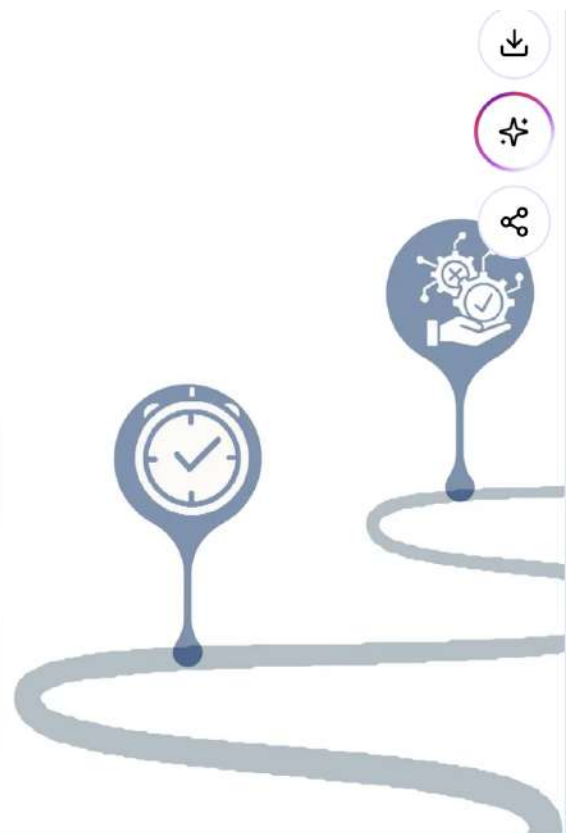
Efficiency and Compliance: The system requires less computing power and ensures compliance with CIS standards, even on standalone, air-gapped systems using up-to-date RAG status indicators.

POTENTIAL CHALLENGES

- **Administrative Access Required:** Dependence on admin rights could limit deployment across environments with restricted access.
- **Integration Complexity:** Integrating with diverse and potentially outdated systems could complicate seamless operation and compatibility.

STRATEGIES TO OVERCOME CHALLENGES

- **Role-Based Access Control Implementation:** An RBAC system can efficiently manage access, ensuring system integrity and security.
- **Robust Testing and Configuration Management:** A thorough testing suite and configuration database help manage system updates and minimize integration issues.





IMPACT AND BENEFITS



IMPACT

- Enhanced Security Compliance
- Scalability and Efficiency
- Reduced Human Error

Error

BENEFITS

- Streamlined Management Process
- Enhanced System Security
- Compliance and Cost
- Efficiency Power of AI without Internet

FOR INSTANCE: USER STORY

Scenario

- A government agency, operating in a highly secure environment with no internet access, adopts our system for managing and deploying GPOs across its network of standalone, air-gapped computers.

Impact

- **Increased Security Compliance:** The agency benefits from the system's 100% offline capability, eliminating the risk of external breaches and enhancing the security of sensitive data.

- **Operational Efficiency:** The agency uses less computing resources and benefits from an up-to-date, error-free configuration due to the system's efficient RAG updates and robust in-house testing.

Benefits

- **Improved GPO Management:** With the ability to save and recall past configurations, the agency streamlines its process for testing and deploying new policies, reducing downtime and operational disruptions.

Cultural Change: The adoption of an RBAC system ensures that only authorized personnel can access critical system settings, promoting a culture of accountability and meticulous access control within the agency.

RESEARCH AND REFERENCES



 [RAG](#)

 [Mistral](#)

 [Ollama](#)

 [CIS
Benchmark](#)

 [Powershell module](#)

 [Air gapped systems](#)

 [Gpo Overview](#)

 [SSH Tunneling](#)